Book construction: Desk book

Portfolio manager report · Book construction · 10 November 2026

Proposed: Crypto Thin Losers and SPY Volatility Managed, money allowed

Abstract

Crypto Thin Losers is a reversal strategy in crypto: it holds thin coins against liquid ones. ETF TSMOM 12-Month is a twelve-month trend strategy on a multi-asset set of ETFs. SPY Volatility Managed holds SPY and scales it down when volatility is high.

On the common window (2021-01-29 to 2026-09-30, 2071 days) the book of all three has a Sharpe of 1.017 at 10.7 % volatility, against 0.49 to 0.76 for the strategies alone, and it is positive in every year of the yearly table. Its Sharpe is above zero in 99.7 % of histories. The IDM is 1.56, so the three behave like about one and a half independent bets, not three. Most of what the book earns on a normal week is equity beta, carried by SPY Volatility Managed, with a crypto sleeve that has its own alpha. ETF TSMOM 12-Month does not earn its place: dropping it hurts in only 46 % of histories.

Proposal: Crypto Thin Losers and SPY Volatility Managed, money allowed by the script's rule; ETF TSMOM 12-Month stays in the library. The book's main need is a source of return that does not depend on equities.

1 The strategies

Each strategy is sized alone at 10 % volatility, then the book gives equal risk by family (trend, reversal, low-vol), applies the IDM of 1.56 and aims at a 10 % target. With three families each gets 33.3 % of the risk.

Crypto Thin Losers has a Sharpe of 0.70 and a t of 2.19, with a maximum drawdown of −15.3 %. Its Sharpe after the luck haircut is also 0.70, because the haircut is 0.00 here. ETF TSMOM 12-Month has a Sharpe of 0.49, a t of 2.16 and a maximum drawdown of −21.3 %, the deepest of the three. SPY Volatility Managed has the best figures, a Sharpe of 0.76, a t of 3.33 and a drawdown of −12.1 %.

Strategy	Family	Sharpe	After luck	t	Max DD	Worst month	Unseen from
Crypto Thin Losers	reversal	0.70	0.70	2.19	−15.3 %	−8.0 %	Jan 2025
ETF TSMOM 12-Month	trend	0.49	0.49	2.16	−21.3 %	−4.5 %	Jan 2021
SPY Volatility Managed	low-vol	0.76	0.76	3.33	−12.1 %	−5.0 %	Jan 2021

Table 1: The strategies, each alone at 10 % volatility on the common window, net. After luck: minus the luck line of its family and asset (Trials summary). t: Newey-West, the strategy's full sample. Unseen from: start of its sealed test. In the book each family takes an equal share of the risk.

2 The book against its strategies

The book's Sharpe of 1.017 is above each strategy alone, and the book beats Crypto Thin Losers in 80 % of histories, ETF TSMOM 12-Month in 96 % and SPY Volatility Managed in 78 %. The 5th to 95th percentile range of the book's Sharpe is wide, 0.304 to 1.626, so the point value should not be read as a promise. The probability of a Sharpe above zero is 99.7 %.

The book's maximum drawdown is −17.2 % (range −26.1 % to −9.4 %), and its worst month is −5.1 %. The worst month of the strategies alone goes from −4.5 % to −8.0 %, so the book

smooths the worst month a little but is not a shelter: the maximum drawdown sits between the best and the worst strategy alone.

Out of sample, from 2025-01-01 (638 days), where every strategy is in its sealed test, the book has a Sharpe of 0.96 and a maximum drawdown of −15.6 %, against 1.05 before. The loss is small. Alone, the strategies were at 0.54 (Crypto Thin Losers), 0.83 (ETF TSMOM 12-Month) and 0.59 (SPY Volatility Managed), all weaker than the book.

	Sample	90 % range
Annualised return	10.9 %	3.4 % to 17.2 %
Annualised volatility	10.7 %	10.0 % to 11.3 %
Sharpe ratio	1.02	0.30 to 1.63
Maximum drawdown	−17.2 %	−26.1 % to −9.4 %
Worst month	−5.1 %	−9.7 % to −4.2 %
Daily CVaR 5 %	−1.34 %	−1.48 % to −1.21 %
Positive months	68 %	54 % to 72 %

Table 2: The book, net, 2021-01-29 to 2026-09-30, at its 10 % target (IDM 1.56). Range: 5th to 95th percentile of a stationary block bootstrap (blocks of 20 days, 2 000 paths).

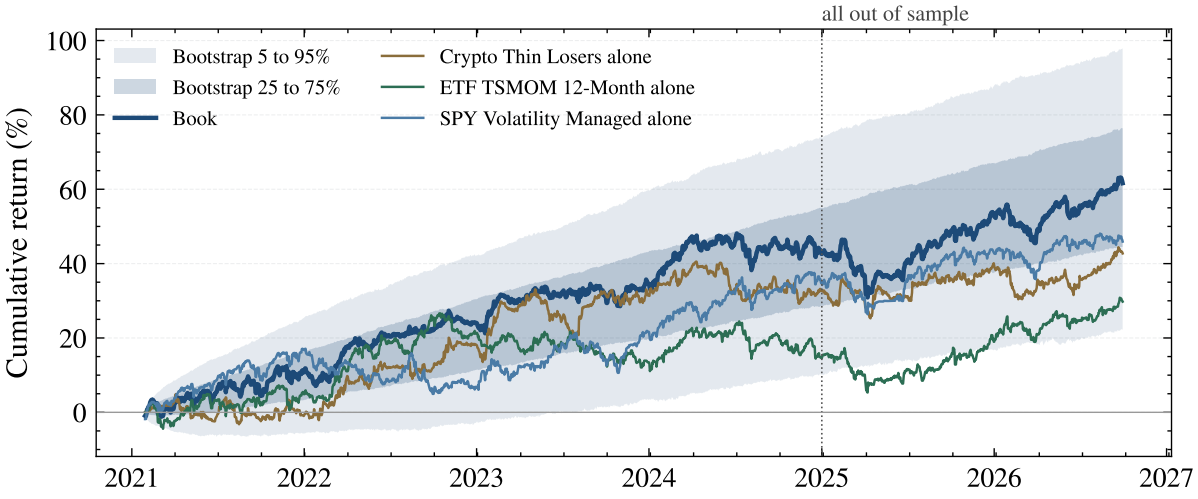


Figure 1: Cumulative net return of the book and of each strategy alone, all at 10 % volatility, with the bands of the book's bootstrap histories. Dotted line: from here every strategy is in its sealed test. Linear scale.

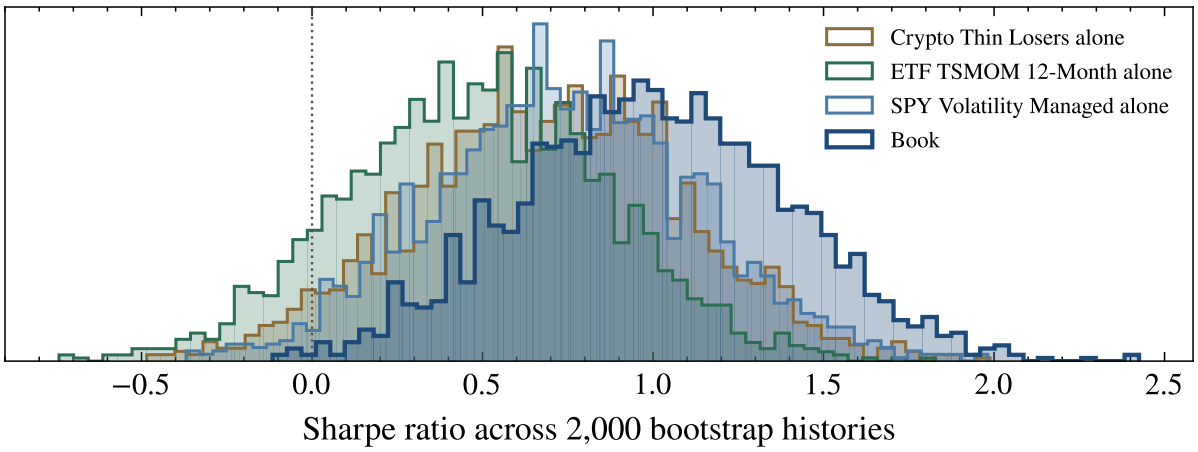


Figure 2: Sharpe ratio of the book and of each strategy alone across 2 000 bootstrap histories.

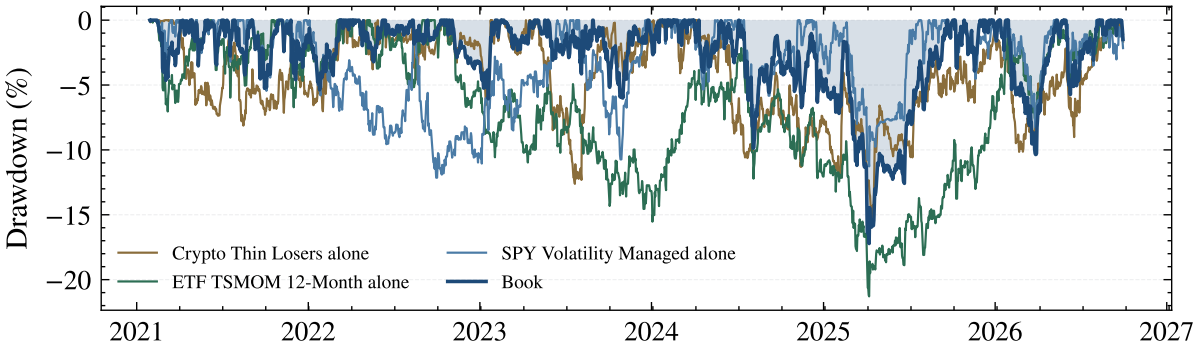


Figure 3: Drawdown of the book (shaded) and of each strategy alone.

	Book	Crypto Thin Losers	ETF TSMOM 12-Month	SPY Volatility Managed
Sharpe, every strategy unseen	0.96	0.54	0.83	0.59

Table 3: Out of sample: from 2025-01-01, when every strategy is in its sealed test (638 days).

3 Does each strategy earn its place?

Without Crypto Thin Losers the book's Sharpe falls to 0.75 and the IDM to 1.19; dropping it hurts in 90 % of histories. This is the strategy that earns the most, because its correlation to the other two is about zero (−0.05 and −0.02).

Without SPY Volatility Managed the Sharpe is 0.87, the maximum drawdown worsens to −21.9 %, and dropping it hurts in 76 % of histories. It earns its place, though less clearly than the crypto strategy, because it is correlated to ETF TSMOM 12-Month (+0.41).

Without ETF TSMOM 12-Month the Sharpe rises to 1.04, the maximum drawdown shrinks to −13.6 %, and dropping it hurts in only 46 % of histories, under the 60 % bar. What it still brings is the equity and bond bear market of 2022, where it made 11.1 %, and a smaller worst month (the book without it has −6.6 %, with it −5.1 %). That is one episode, and the strategy lost in 2023 and gave a negative alpha. It is not enough for a place now.

Book	Sharpe	IDM	Drop hurts	Max DD	Worst month	Beats alone
All strategies	1.02	1.56		−17.2 %	−5.1 %	
Without Crypto Thin Losers	0.75	1.19	90 %	−17.1 %	−4.8 %	80 %
Without ETF TSMOM 12-Month	1.04	1.43	46 %	−13.6 %	−6.6 %	96 %
Without SPY Volatility Managed	0.87	1.45	76 %	−21.9 %	−4.6 %	78 %

Table 4: Does each strategy earn its place? Drop hurts: share of bootstrap histories in which the book of all strategies has a higher Sharpe ratio than the book without that one. Beats alone: share in which the book's Sharpe ratio is above that strategy's alone.

4 Diversification over time

The correlations between strategies are low: Crypto Thin Losers is at −0.05 and −0.02 to the others, and only ETF TSMOM 12-Month with SPY Volatility Managed is at +0.41. The IDM of 1.56 is therefore well below 3, the value of three independent bets.

The 6-month correlation of Crypto Thin Losers to the rest stays between −0.22 (2026-01-30) and +0.28 (2023-02-16), last −0.12. For ETF TSMOM 12-Month it swings from −0.58 (2022-11-02) to +0.68 (2022-02-20), last +0.34. For SPY Volatility Managed it goes from −0.47 (2023-01-06) to +0.67 (2021-12-13), last +0.41. The average pairwise correlation peaked at +0.33 on 2025-05-13, lowest −0.24, last +0.14. The two stable legs are crypto and equity; the trend leg is the unstable one.

Risk shares are balanced on average (31 %, 33 % and 36 %) but the crypto share ranges from 11 % to 71 %, and the trend share from 2 % to 49 %. Return shares are 36 % for Crypto Thin Losers, 25 % for ETF TSMOM 12-Month and 39 % for SPY Volatility Managed.

One suspect day is listed: 2023-08-06, Crypto Thin Losers at 8.2 % in a day, beyond 8 standard deviations. It inflates that strategy's 2023 figure and its Sharpe a little. It must be checked in the runs; I did not touch the data.

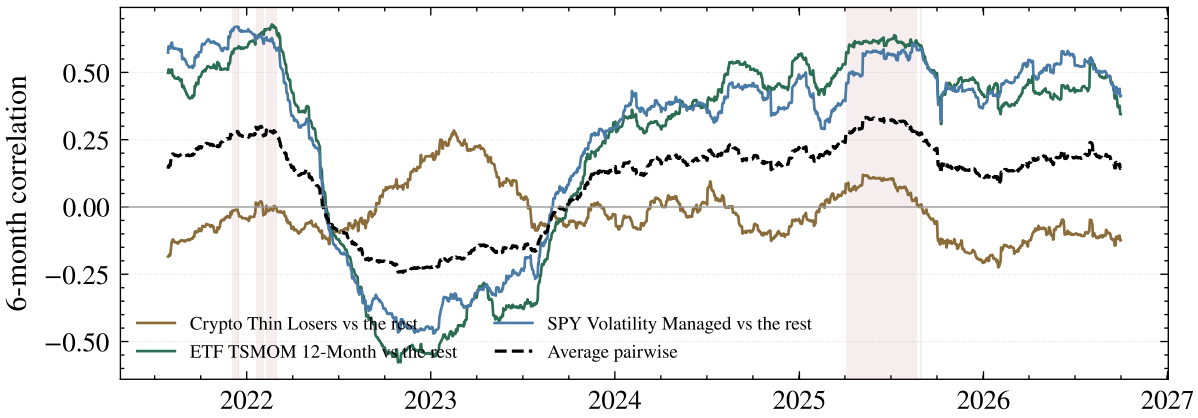


Figure 4: Six-month rolling correlation of each strategy with the rest of the book, and the average pairwise correlation (dashed). Shaded: the 10 % of days when the average was highest.

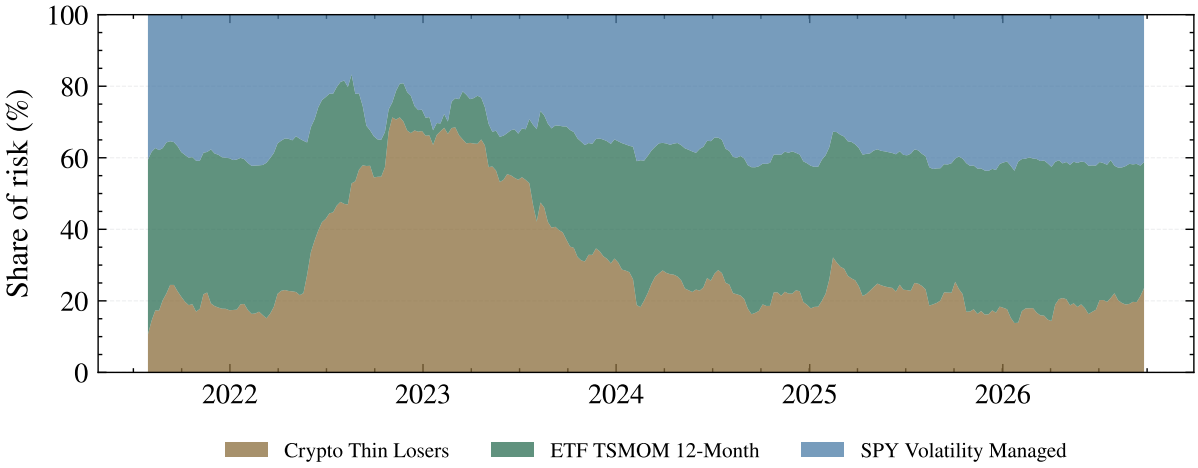


Figure 5: Share of the book's risk carried by each strategy, six-month rolling covariance.

5 Hidden exposures

SPY Volatility Managed is a market in disguise: its beta to SPY is +0.54 (t 7.7), its R2 is 0.85 and its alpha is −0.4 %/yr (t −0.2). It earns the equity premium with a volatility filter and nothing else. Crypto Thin Losers is the opposite: alpha 8.4 %/yr (t 2.5), R2 0.05, no significant beta. ETF TSMOM 12-Month has an alpha of −3.0 %/yr (t −0.9), a R2 of 0.16, a short TLT position (−0.22, t −2.4) and a long gold position (+0.18, t 2.6).

The book has an alpha of 2.6 %/yr (t 0.8), a SPY beta of +0.38 (t 3.7) and a R2 of 0.29. The alpha is not significant on its own; it comes from the crypto sleeve.

The book's 26-week correlation to SPY is +0.58 over the full window, from −0.16 (2023-04-30) to +0.90 (2026-01-18), last +0.79, above 0.3 in absolute value in 83 % of weeks. The correlation to BTC is +0.27, from −0.52 (2024-03-24) to +0.67 (2022-05-29), last +0.30, above 0.3 in 58 % of weeks. Most of the time this is an equity book with a market-neutral crypto sleeve, and the equity link is strong right now.

	Alpha	t	SPY	TLT	UUP	GLD	BTC	R^2
Crypto Thin Losers	8.4 %	2.5	−0.04	−0.10	−0.17	−0.08	0.03	0.05
ETF TSMOM 12-Month	−3.0 %	−0.9	0.23	**−0.22**	0.32	**0.18**	−0.04	0.16
SPY Volatility Managed	−0.4 %	−0.2	**0.54**	**0.11**	0.09	0.00	0.02	0.85
Book	2.6 %	0.8	**0.38**	−0.11	0.13	0.05	0.00	0.29

Table 5: Monthly net returns regressed on the markets at once: US equities (SPY), long Treasuries (TLT), the dollar (UUP), gold (GLD), bitcoin. At 10 % volatility. Alpha annualised, Newey-West t (3 lags). Betas in bold: $|t| \geq 2$.

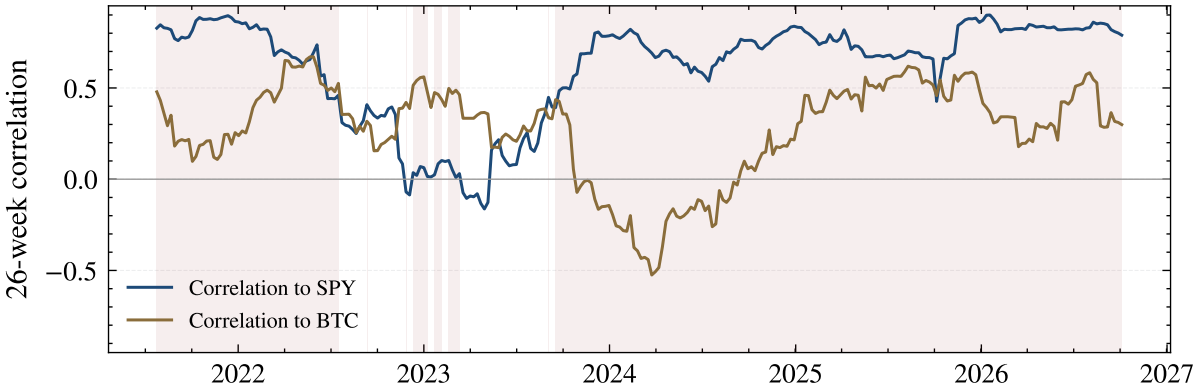


Figure 6: 26-week rolling correlation of the book to US equities and to bitcoin, weekly returns. Shaded: either correlation above 0.4 in absolute value.

6 Stress episodes

In slow crises the book holds. It made 12.3 % in the equity and bond bear market of 2022, carried by ETF TSMOM 12-Month (11.1 %) and Crypto Thin Losers (7.2 %), while SPY Volatility Managed lost 6.0 %. It made 2.5 % during Terra and 3AC and 3.3 % in the China ban.

It is the fast crashes that cost. The yen carry unwind lost 4.1 % in a few days, with ETF TSMOM 12-Month at −2.5 % and SPY Volatility Managed at −2.0 %. The tariff shock lost 6.0 %, the worst episode, with SPY at −12 %; SPY Volatility Managed lost 2.2 %, ETF TSMOM 12-Month 2.5 % and Crypto Thin Losers 1.3 %. FTX (−1.0 %) and the regional banks (−1.1 %) were small. In the fast episodes all three legs lose together, which is what the equity link predicts.

Episode	Dates	SPY	BTC	Book	Crypto Thin Losers	ETF TSMOM 12-Month	SPY Volatility Managed
Crypto crash, China ban	May 21–Jul 21	7 %	−40 %	3.3 %	2.7 %	−1.0 %	1.6 %
Equity and bond bear market	Jan 22–Oct 22	−24 %	−59 %	12.3 %	7.2 %	11.1 %	−6.0 %
Terra and 3AC	May 22–Jun 22	−11 %	−48 %	2.5 %	1.9 %	2.7 %	−2.1 %
FTX collapse	Nov 22	4 %	−25 %	−1.0 %	−0.2 %	−1.4 %	0.6 %
US regional banks	Mar 23	−2 %	26 %	−1.1 %	−0.4 %	−0.1 %	−0.6 %
Yen carry unwind	Jul 24–Aug 24	−6 %	−16 %	−4.1 %	0.4 %	−2.5 %	−2.0 %
US tariff shock	Apr 25	−12 %	−8 %	−6.0 %	−1.3 %	−2.5 %	−2.2 %

Table 6: Stress episodes. SPY and BTC: price change over the episode. Book: its cumulative return; then each strategy's contribution to it.

7 Proposed book

Two rules apply. A strategy earns its place if dropping it hurts in 60 % of histories or more: Crypto Thin Losers (90 %) and SPY Volatility Managed (76 %) pass, ETF TSMOM 12-Month (46 %) does not. Money needs status passed, t of 2 or more, and a Sharpe after luck above zero: the two kept strategies meet it.

The proposed book of Crypto Thin Losers and SPY Volatility Managed has a Sharpe of 1.04, a volatility of 10.7 %, a maximum drawdown of −13.6 %, a worst month of −6.6 % and an IDM of

1.43. It beats the book of all three in 54 % of histories, which is close to a coin toss: the gain from dropping the trend leg is small, and the choice rests on the rule, not on a clear win.

The script's decision is: Crypto Thin Losers and SPY Volatility Managed, money allowed. I add a warning that the script cannot see. The Test notes of both strategies say paper only, no money (crypto: ambiguous zone, the mechanism did not pay on the long leg; SPY: final proof t 1.95 against a bar of 2). I do not upgrade the decision and I do not downgrade it, but the user should read "money allowed" next to those two verdicts. Paper means tracking the book month by month without capital. What would change the proposal: a longer track for the crypto strategy, or a stronger trend candidate that earns a place.

Strategy	Drop hurts	After luck	t	Proposed
Crypto Thin Losers	90 %	0.70	2.19	in the book, money allowed
ETF TSMOM 12-Month	46 %	0.49	2.16	out, in the library
SPY Volatility Managed	76 %	0.76	3.33	in the book, money allowed
Proposed book		**Sharpe 1.04, max DD −13.6 %, money**		

Table 7: The proposal. A strategy earns its place if dropping it hurts the book in 60 % of histories or more; it can carry money with status passed, a full-sample t of 2 or more and a Sharpe ratio after luck above zero. The book carries money only if all its members can.

8 What the book needs

The book lacks a return source that does not depend on equities or on crypto reversal. It is missing rates, commodities and currency premia, and anything that pays in a fast equity sell-off, since the three legs lost together in the yen unwind and the tariff shock. The Ideator should look for low-equity-beta strategies, ideally with a positive result in sharp drawdowns.

9 Limits

The window is short, 69 months, and holds a single market cycle with one bear year. Weights are equal by family on purpose: optimised weights would fit noise. The leave-one-out comparison is made on the same years as the selection, so the 46 % for ETF TSMOM 12-Month is not far from the 60 % bar and could move on another window. Each strategy is partly in its training and validation years, so the full-window Sharpe is not all out of sample. The suspect day of 2023-08-06 in Crypto Thin Losers is unchecked. The markets regression uses 69 monthly points, so the betas are rough. No strategy here was imported from outside the desk.

A Returns by year

	Book	Crypto Thin Losers	ETF TSMOM 12-Month	SPY Volatility Managed
2021	10.7 %	−1.0 %	4.9 %	16.7 %
2022	13.4 %	19.2 %	16.5 %	−10.0 %
2023	10.5 %	13.2 %	−7.9 %	14.9 %
2024	8.1 %	1.3 %	1.1 %	13.1 %
2025	10.4 %	7.4 %	4.4 %	8.1 %
2026	8.6 %	2.6 %	10.8 %	3.2 %

Table 8: Net return by year, the book and each strategy alone at 10 % volatility.

B Correlation matrix

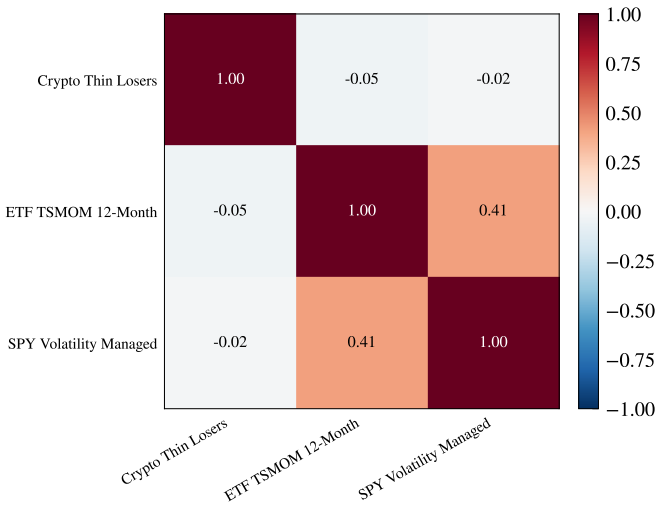


Figure 7: Daily correlation of net returns over the whole window.

C Sources

Strategy	Source of the returns
Crypto Thin Losers	runs 20261024-083824-train-base, 20261023-164211-validation-base, 20261024-083811-test-base
ETF TSMOM 12-Month	runs 20261109-091945-train-base, 20261109-103847-validation-base, 20261109-103855-train+validation-base, 20261109-112907-test-base
SPY Volatility Managed	runs 20261025-142825-train-base, 20261025-160847-train+validation-base, 20261026-085632-test-base
Markets	SPY, TLT, UUP, GLD adjusted close; BTCUSDT mark price
Sizing	lab/book_report.py (construct)

Table 9: Sources.